Teaching notes and solutions

> **Teacher note · Teacher notes.** Test 1 comes before S12 and S15: set vector geometry in a given basis, without requiring new coordinate techniques. Use a physical cube, then an oblique perspective drawing; mark the hidden edges. Two non-parallel lines in space do not necessarily intersect. The norm via coordinates requires an orthonormal frame. Formal parametric representations and equations of planes will be covered later.

Solutions to the activity

1. By Chasles, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{\jmath}$ and $\overrightarrow{IK} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{k}$.

2. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would be $\frac{1}{2} = 0$, impossible in a basis. These two vectors are therefore non-collinear; I, J, K define a plane.

3. The centre of the cube satisfies $\overrightarrow{AO} = \frac{1}{2}(\vec{\imath} + \vec{\jmath} + \vec{k})$, so $\overrightarrow{IO} = \frac{1}{2}\vec{\jmath} + \frac{1}{2}\vec{k}$. If $O \in (IJK)$, there exist s, t such that $\overrightarrow{IO} = s\overrightarrow{IJ} + t\overrightarrow{IK}$. Comparing the coefficients of the basis,

$$0 = -\frac{s+t}{2}, \qquad \frac{1}{2} = \frac{s}{2}, \qquad \frac{1}{2} = \frac{t}{2}.$$

   The last two force $s = t = 1$, whereas the first would require $s + t = 0$. Contradiction: $O \notin (IJK)$.

4. Oblique perspective is a projection onto a sheet of paper: distinct points may have the same image. It helps to conjecture, but an apparent overlap does not prove membership of a plane.

E1. In the cuboid, $\overrightarrow{BC} = \overrightarrow{AD}$ and $\overrightarrow{CG} = \overrightarrow{AE}$, so $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{AD} + \overrightarrow{AE}$ and $G(1,1,1)$. The centre O is the midpoint of $[AG]$: $O(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. The frame is not orthonormal: its basis vectors are orthogonal, but their norms are 4, 3 and 2. Lengths therefore cannot be read directly from the coordinates.

E2. The midpoint has coordinates $(\frac{0+4}{2}, \frac{1-1}{2}, \frac{2+0}{2}) = (2,0,1)$. The coordinate differences give $\overrightarrow{AB} = (4,-2,-2)$; in the orthonormal frame, $AB = \sqrt{16+4+4} = 2\sqrt{6}$.

E3. We calculate $2\vec{u} + 3\vec{v} = (2,0,2) + (0,3,3) = (2,3,5) = \vec{w}$. Thus $\vec{w}$ belongs to the vector plane spanned by the first two vectors: the three are coplanar.

E4. The first two vectors are non-collinear. If the third were their combination $\lambda(1,0,1) + \mu(0,1,1)$, the first two coordinates would force $\lambda = \mu = 1$, but the third would equal $2 \neq 0$. The three vectors are non-coplanar; they therefore form a basis of space.

E5. $\overrightarrow{AB} = (2,-1,2)$ and $\overrightarrow{AC} = (4,-2,4) = 2\overrightarrow{AB}$. Since $A \neq B$, the line (AB) is defined and this collinearity proves $C \in (AB)$.

Solutions to Homework 10

1. By Chasles and the definitions of midpoints,
$$\overrightarrow{IJ} = \tfrac{1}{2}\overrightarrow{AC} - \tfrac{1}{2}\overrightarrow{AB} = \tfrac{1}{2}\overrightarrow{BC}, \qquad \overrightarrow{IK} = \tfrac{1}{2}\overrightarrow{AD} - \tfrac{1}{2}\overrightarrow{AB} = \tfrac{1}{2}\overrightarrow{BD}.$$

2. The points B, C, D are not collinear, since the tetrahedron is non-degenerate. The vectors $\overrightarrow{BC}, \overrightarrow{BD}$ are therefore non-collinear, and so are $\overrightarrow{IJ}, \overrightarrow{IK}$. The planes have the same direction. They are distinct: if $I \in (BCD)$, the line $(BI) = (BA)$ would be contained in this plane, which would place A in it, a contradiction. They are therefore strictly parallel.

3. In the basis $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, we have $\overrightarrow{IG} = (-\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\overrightarrow{IJ} = (-\frac{1}{2}, \frac{1}{2}, 0)$ and $\overrightarrow{IK} = (-\frac{1}{2}, 0, \frac{1}{2})$. A relation $\overrightarrow{IG} = s\overrightarrow{IJ} + t\overrightarrow{IK}$ forces $s = t = \frac{1}{2}$ by the last two coordinates; the first would then equal $-\frac{1}{2}$, and not $-\frac{1}{4}$. Hence $G \notin (IJK)$.

4. In this particular case, $I(2,0,0)$, $J(0,2,0)$ and $K(0,0,2)$. We find $IJ = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$, and likewise $IK = JK = 2\sqrt{2}$. The triangle is equilateral. This metric conclusion does not hold for every non-degenerate tetrahedron: it uses the coordinates of the particular case.

Solutions to the textbook or past-paper work

In this exercise, I is the midpoint of $[AB]$ and J that of $[CD]$.

1. The planes (AIE) and (BIG) both contain (AB). They are distinct: G does not belong to the plane of the face $ABFE$. Their intersection is therefore exactly the line (AB).

2. The points A, D, I are three non-collinear points of the face $ABCD$; the same holds for B, J, C. The two planes are therefore both equal to (ABC). The intersection is the whole plane, not a line.

3. The plane (HEF) is that of the top face; (BJC) is that of the bottom face. These planes are strictly parallel: their intersection is empty.